

Metaphysical Foundations and Triangulation Invariants of the Neurosymbolic Forest

System Architecture Invariant Manifest

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1 Introduction and Ontological Modalities

Following the formalisms of Edward Zalta's Abstract Object Theory and Situation Theory, we isolate three distinct modal world-lines within a single unified processing manifold:

1. **The Historical Immutable World:** Rigidly designated actualities (s_{actual}) frozen in the past (e.g., Moses, David, Paul). They are structurally locked against retroactive modifications.
2. **The Fictional Mutable World:** Abstract objects encoding properties inside an isolated situation domain (s'), completely open to counterfactual planning rewrites (e.g., John, Judith, Pierre).
3. **The Indexical Real World:** The concrete evaluation layer where sovereign actions and executions occur in real time.

2 The Core Logic of the Unified Background Knowledge Base (kb.pl)

Rather than forcing individual decision trees to duplicate rules, our architecture centralizes all invariant logical axioms inside a single background knowledge layer: `kb.pl`. This provides the absolute mathematical framework that governs the entire neurosymbolic universe across three distinct axes:

2.1 Graph Minor Transformations

The knowledge base specifies the structural rules for graph path validation. When a link is severed, the platform computes path clearance over the Grigorchuk Infinite Group Matrix by traversing equivalent vertex steps:

$$\text{TotalCost} = \sum \text{EqCost} + \text{StepCost} + \text{SubCost} \quad (1)$$

This allows the system to evaluate non-linear journey repairs symmetrically.

2.2 Pragmatic Contract Governance

`kb.pl` handles second-order social contract permissions. Given an action p and a mandatory authorization constraint q , an unauthorized exception state triggers an immediate structural breach vertex:

$$\text{Pragmatic_Breach}(p, q) \iff p \wedge \neg q \quad (2)$$

2.3 Analogy Fact Transfer Laws

The centralized engine hosts the rules of logical transfer:

1. **Reflexive Rule:** Symmetrically infers facts between separate domains ($S_1 \neq S_2$) if an underlying structural match is satisfied.
2. **Transitive Rule:** Chains recovery metrics across multi-point domains ($S_1 \rightarrow S_2 \rightarrow S_3$) by tracing back to the root fact origin point.

3 The Principle of Triangulation via Channel Infomorphisms

While Jon Barwise formulated **Channel Composition and Infomorphisms** to track how information commutes through distributed networks, we deploy the **Principle of Triangulation** to secure cross-model analogies.

An individual decision tree cannot verify a robust analogy on its own; it merely reaches a terminal node. However, if three completely independent structural representations of the same story all arrive at the exact same leaf category, the case for the analogy is mathematically finalized.

3.1 Dimension I: FCA Parthood Ontologies (Tree #6)

Using Formal Concept Analysis (FCA), multi-sentence narratives are decomposed into an 8-dimensional attribute matrix mapping abstract structural parthood ($\sqsubseteq_{\text{part.of}}$):

$$R_{\text{FCA}} = \langle \text{Syntax, Type, Region, Event, Argument, Action, Set, Time} \rangle \quad (3)$$

3.2 Dimension II: Graph Structural Invariants (Tree #8)

Narratives are unrolled as structural networks, tracking topological invariants to observe structural growth:

$$G_{\text{Metrics}} = \langle \text{Topology Type, Logic Property, Tree Width, Graph Border} \rangle \quad (4)$$

While a macroscopic lifecycle appears as a Directed Acyclic Graph (DAG), adding microscopic situational details (e.g., Moses at the burning bush or authoring Scripture) causes the network to grow dynamically into a cyclic mesh topology.

3.3 Dimension III: Logical Fallacy Patterns (Tree #9)

Structural defects in argumentation are parsed into high-level fallacy classifications based on lexical and logical schemes:

$$F_{\text{Class}} = \langle \text{Ad Hominem, Circular Reasoning, Irrelevant Authority} \rangle \quad (5)$$

4 The Spiritual Homomorphism: Judith and the Prodigal Son

The highest layer of analogical inference uses Tree #7 to map Judith's mountain descent to the spiritual recovery trajectory of the Prodigal Son ($h : \text{Judith} \rightarrow \text{Prodigal.Son}$).

Source Scenario (Judith on Mt. Ateh)	$\xrightarrow{h}$	Target Scenario (The Prodigal Son)
Ascent under ease and abundance	$\longrightarrow$	Initial inheritance and comfortable safety
Decision to take a different unverified route	$\longrightarrow$	Departure to a far, unfamiliar country
Helpful piles of stones completely peter out	$\longrightarrow$	The onset of severe famine and exhaustion
Hopelessly lost and fallen on a scree slope	$\longrightarrow$	Stranded in poverty, feeding swine in the field
Injured and trapped against a hardy thorn bush	$\longrightarrow$	The awakening event: coming to his senses
Remembering and flashing the working flashlight	$\longrightarrow$	The return journey and confession: "I have sinned"
Helicopter searchlight illuminates the face	$\longrightarrow$	The Father running out to embrace and kiss
Waving joyously at the incoming aircraft	$\longrightarrow$	The ultimate feast of celebration and redemption

4.1 Structural Asymmetries

The model tracks boundary mismatches explicitly to evaluate the limits of the comparison:

$$\begin{aligned} \text{Judith}_{\text{Boundary}} &= \langle \text{unknown_post_rescue}, \text{judith_alone} \rangle \\ \text{Prodigal}_{\text{Boundary}} &= \langle \text{celebration_party_gift}, \text{jealous_brother_presence} \rangle \end{aligned}$$

Through Channel Triangulation, the system proves that the transition from a broken fall to an external rescue is governed by an invariant top-down law of redemption, while isolating domain-specific boundary mismatches safely.